# Design and Simulation of a Microcontroller Based Voltage Protector

## Objective and design approach

The assessment asks for a device built around a microcontroller that disconnects an appliance when the supply voltage or load current goes out of range, interface to a computer, and lets the user change its settings. I built it as a Wokwi simulation around an ESP32 and kept the logic plain: sample both inputs, compare them with the limits, then drive the relay, LEDs, display and serial port. Two potentiometers stand in for the voltage and current sensors, so I could sweep each input by hand and watch how the controller reacted. They are not calibrated sensors, and nothing in this project touches mains power.

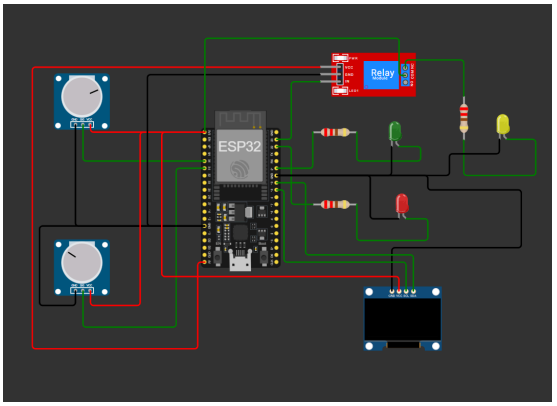



Figure 1. Wokwi circuit of the simulated protection controller.

## Implementation

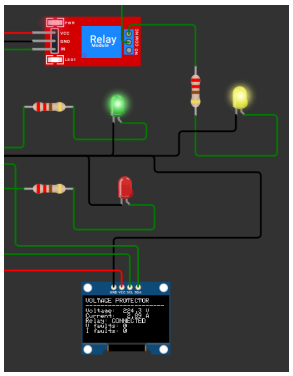
The firmware (src/sketch.ino) reads GPIO 34 for voltage and GPIO 35 for current every 100 ms with the 12 bit ADC. It scales the readings to 0 to 300 V and 0 to 10 A. The default safe windows are 220 to 240 V and 1 to 3 A. When either reading leaves its window, the ESP32 opens the relay on GPIO 23, lights the red LED on GPIO 22 and records which fault occurred. The yellow LED sits on the relay's normally closed contact and stands in for the protected appliance. The load starts disconnected at power up. Once readings are back inside the limits, the controller waits 3 seconds, then reconnects the load and lights the green LED on GPIO 21. A 128 by 64 SSD1306 OLED on I2C (SDA 19, SCL 18) shows voltage, current, relay state and the number of voltage and current fault events.

## Simulation results

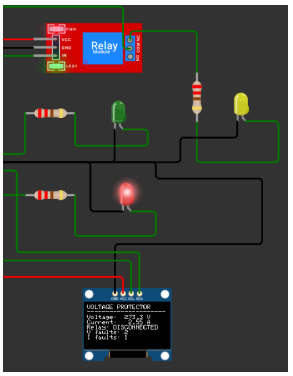
Four scenarios has ran by turning the two potentiometers and recorded what the OLED and LEDs showed. The values below come from the screenshots in test_evidence/. I did not time the response or check the accuracy.

| Scenario | OLED reading | Relay | Faults (V / I) | LEDs |
|---|---|---|---|---|
| **Normal** | 224.3 V, 2.89 A | Connected | 0 / 0 | Green and yellow on |
| **Voltage fault** | 273.3 V, 2.55 A | Disconnected | 2 / 1 | Red on, yellow off |
| **Current fault** | 224.3 V, 5.63 A | Disconnected | 0 / 1 | Red on, yellow off |
| **Combined fault** | 223.3 V, 2.70 A | Disconnected | 2 / 2 | Red on, yellow off |

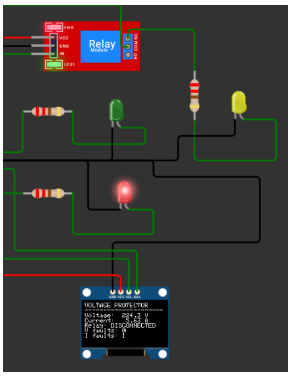
The combined screenshot was taken after both faults had been logged, so the readings on screen are already back inside the limits while the relay is still open.



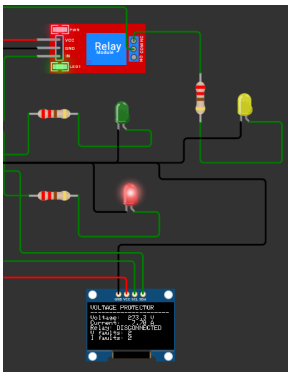
Normal



Voltage fault



Current fault



Combined fault

## Configuration and computer communication

Over the serial monitor at 115200 baud, `HELP` lists the commands and `STATUS` prints the readings, limits, fault state, relay state and event counts. `SET VMIN`, `SET VMAX`, `SET IMIN` and `SET IMAX` change the limits while the program runs, for example `SET VMAX 245`. The firmware rejects a value that would put a minimum above its maximum or go past 300 V or 10 A. It also prints a status line every second and logs each fault and recovery as it happens. The limits are held in RAM, so a reset brings back the defaults.

## Limitations

This is a simulation, so the evidence is qualitative. I have not measured response time, sensing accuracy or relay behavior under load, and the serial commands and the 3 second recovery delay are shown in the code rather than in the screenshots. The current window also treats anything below 1 A as a fault, so a lightly loaded appliance would be disconnected, and that choice needs another look before real use. A physical version would need proper voltage and current sensing with isolation, a relay rated for the load, and bench testing at low voltage first. I would not connect this prototype to mains.

Repository: github.com/tanjimsamad/voltage-protector | Wokwi project: wokwi.com/projects/477395800857832449